Navigating the privacy vs. utility trade-off in network anonymization

Rachel de Jong, Mark van der Loo, Frank Takes

- Universiteit Leiden, Statistics Netherlands

NetSci Maastricht | June 5th 2025

PLANET-NL
Netherlands (NL) Platform for Large-scale Analysis of NETworks

Universiteit van Amsterdam

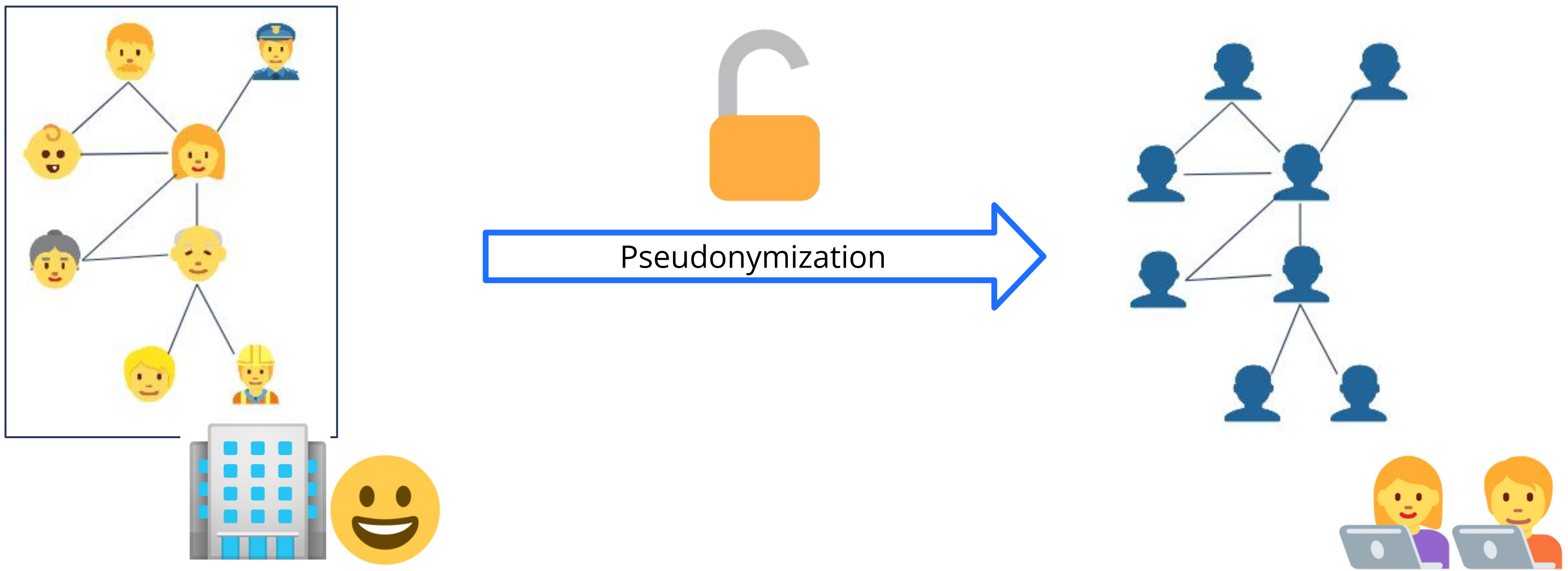
Universiteit Leiden

NWO

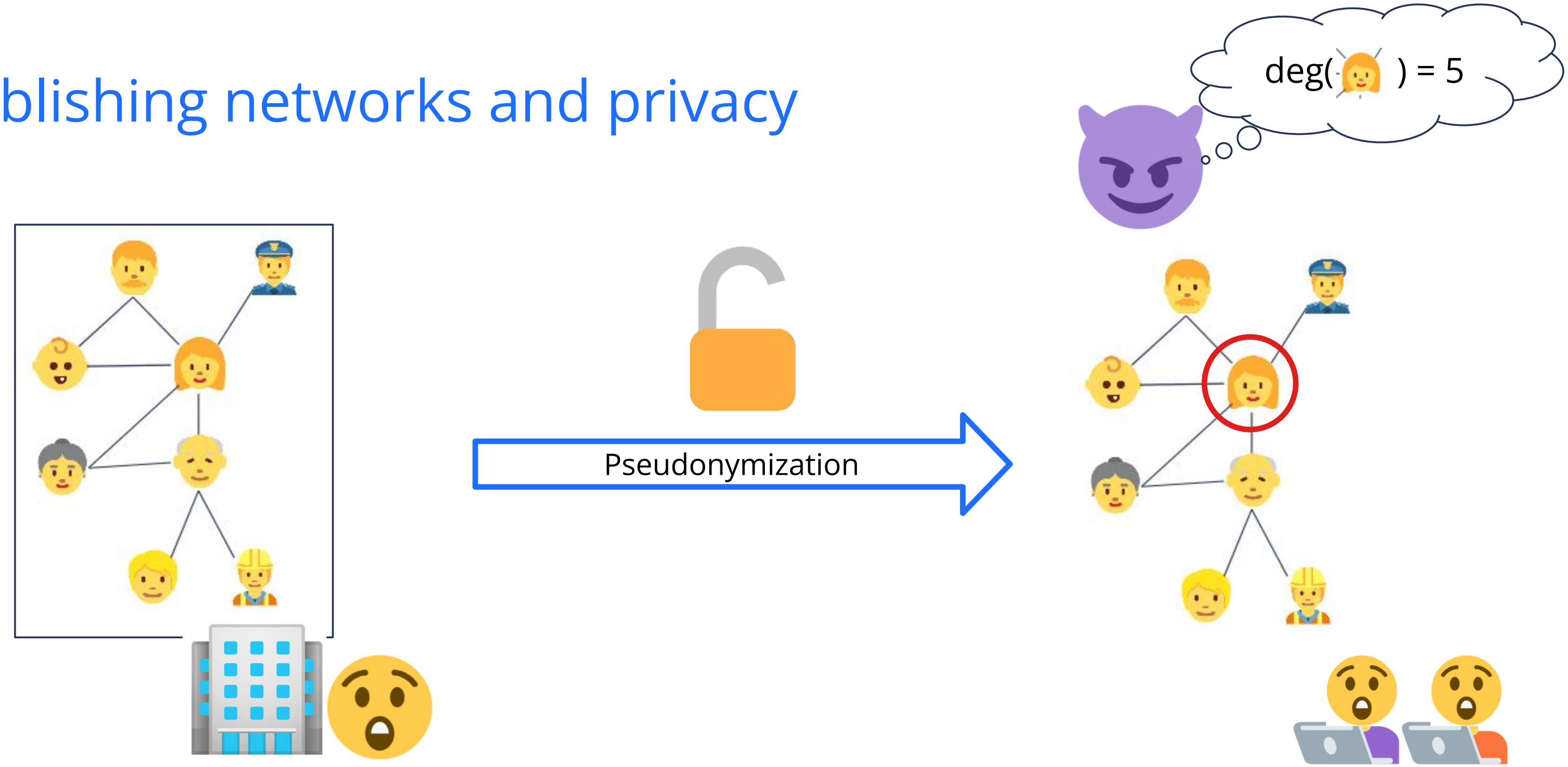
SSHOC NL
social sciences & humanities open cloud

PDI SSH
Platform Digitale Infrastructuur Social Sciences & Humanities

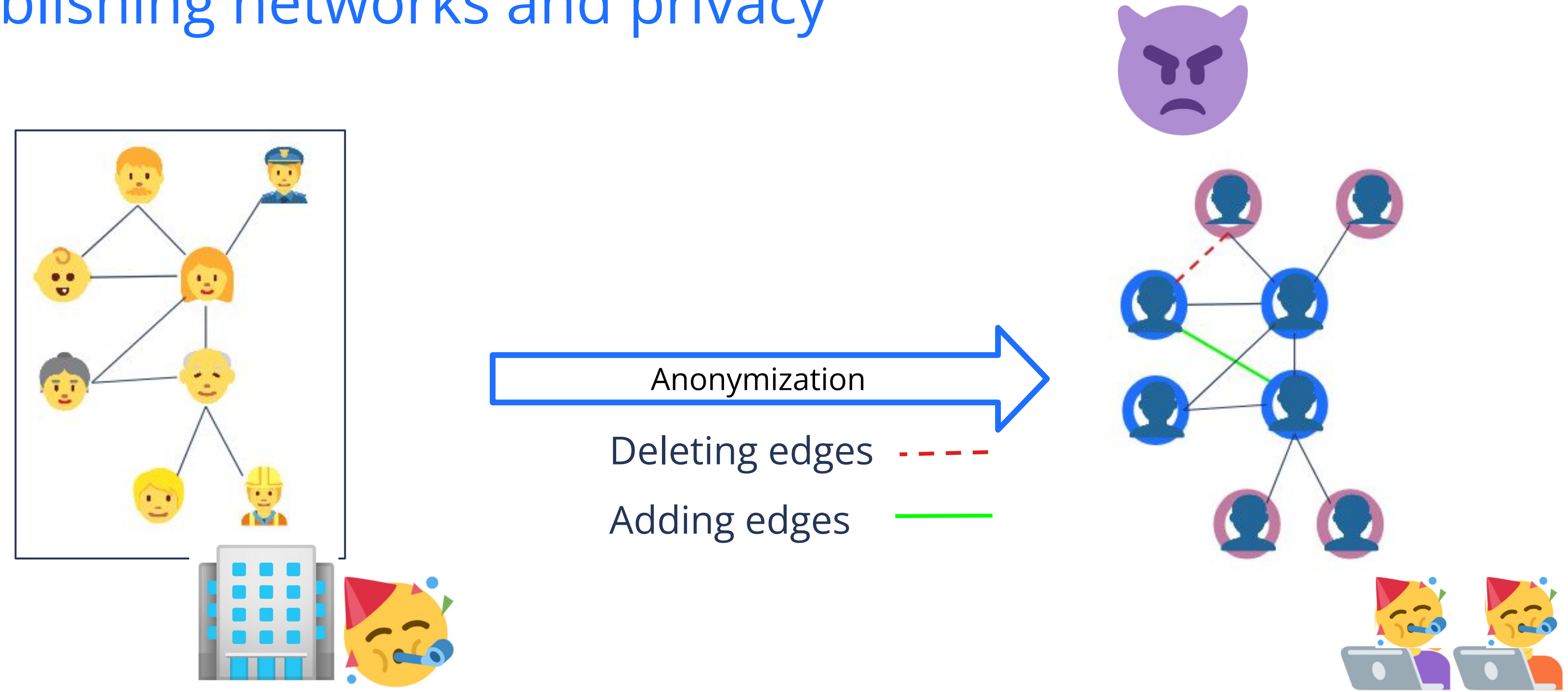
Publishing networks and privacy



Publishing networks and privacy



Publishing networks and privacy



Q1: How to **measure** anonymity?

Q2: How to **anonymize** a network?

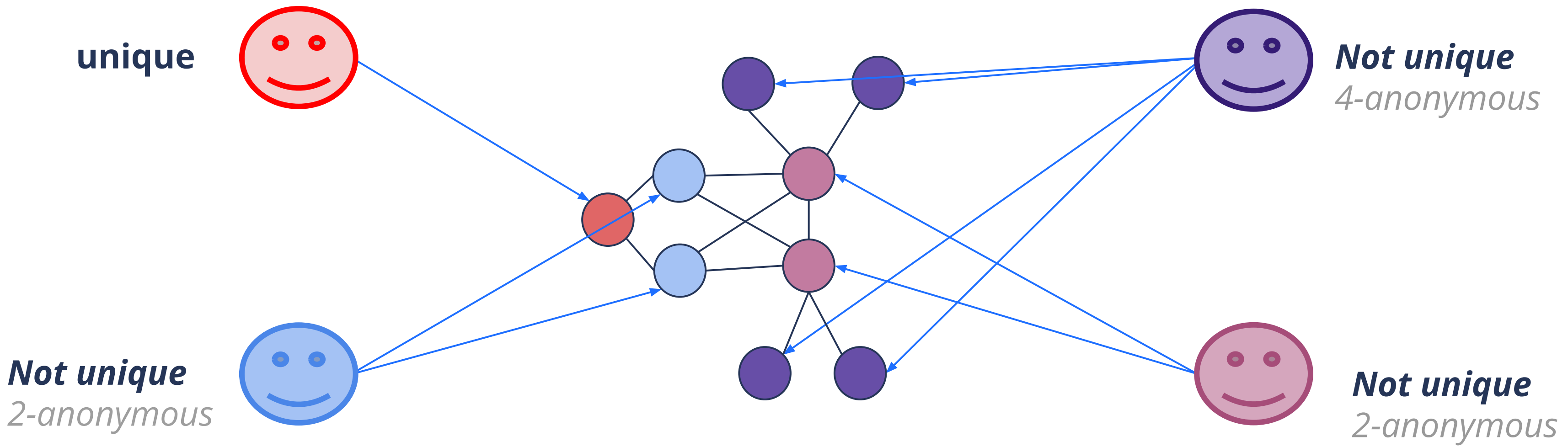
Q3: What is their effect on **utility**?

k-Anonymity

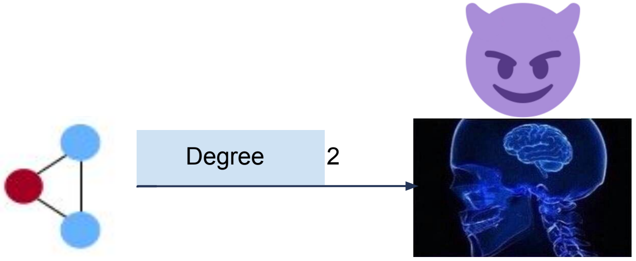
How many candidates are there for each individual?

For now: **Not unique → Anonymous**

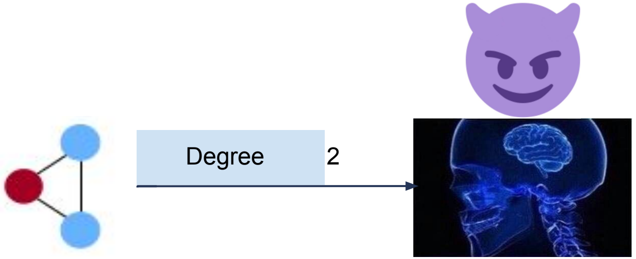
anonymity = uniqueness: fraction of unique nodes

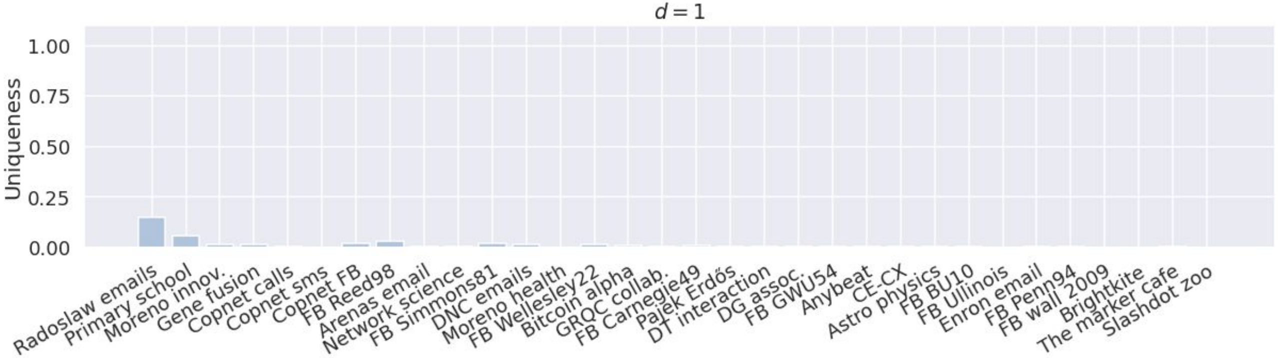


Anonymity measures



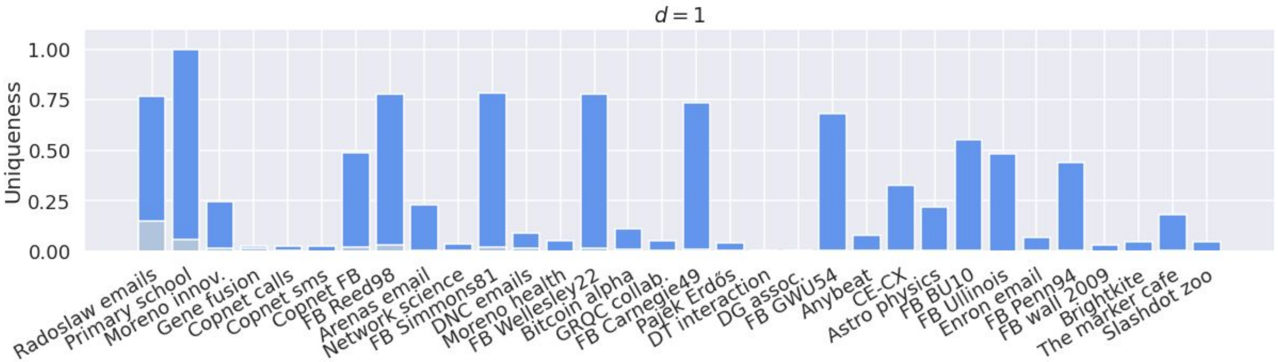
Anonymity measures

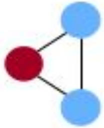




|V|: ~100 - 3M |E|: ~100 - 18M

Anonymity measures

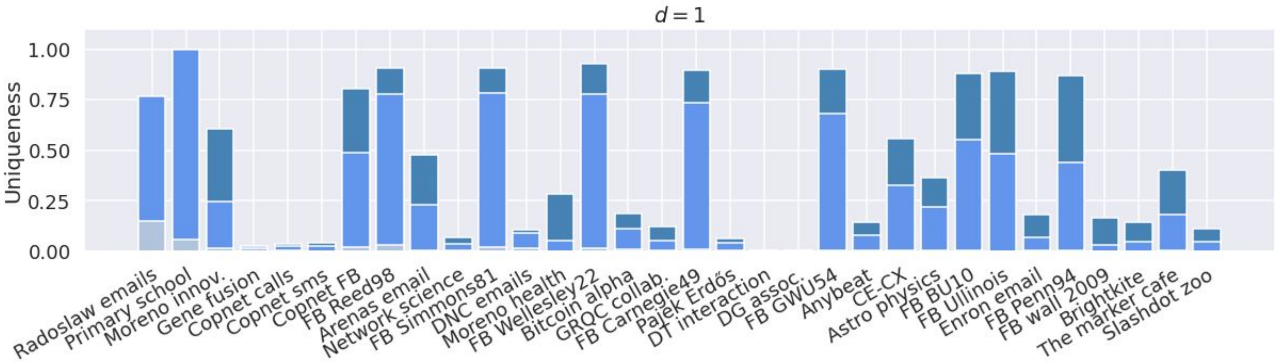


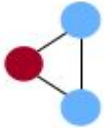


Degree	2
(|V|, |E|)	(3, 3)

|V|: ~100 - 3M |E|: ~100 - 18M

Anonymity measures

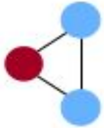


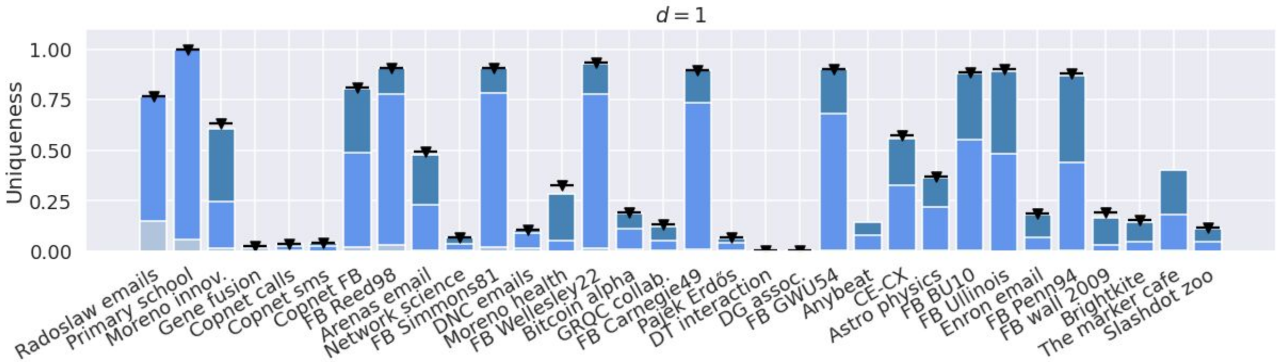


Degree	2
(|V|, |E|)	(3, 3)
Degree distribution	{2, 2, 2}

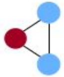
|V|: ~100 - 3M |E|: ~100 - 18M

Anonymity measures

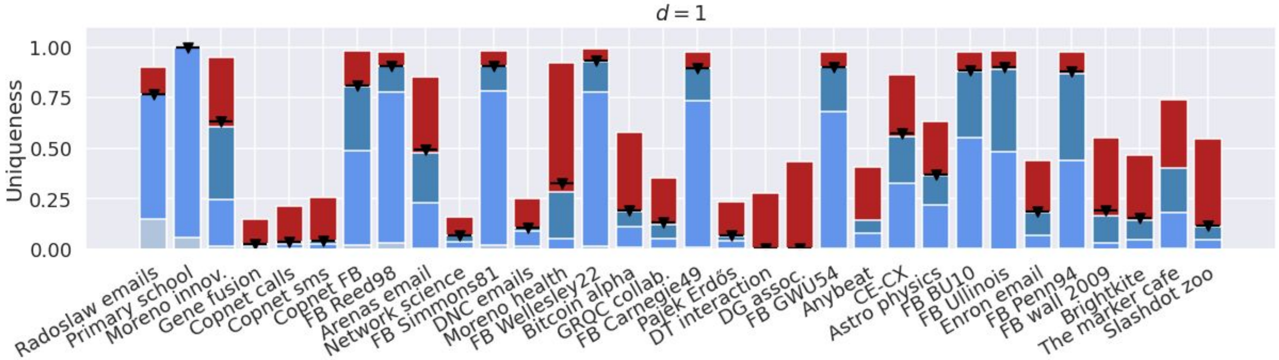




|V|: ~100 - 3M |E|: ~100 - 18M

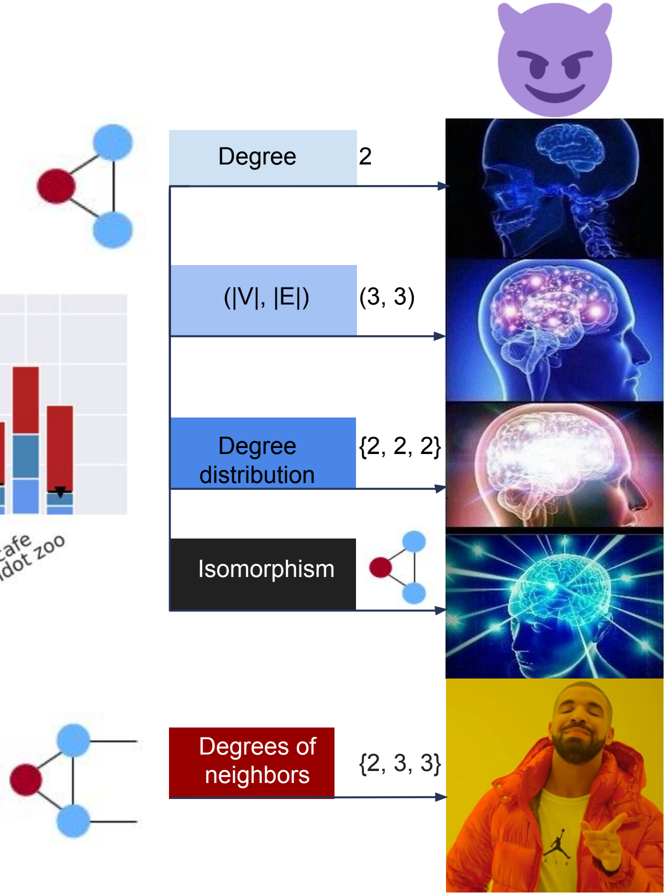
Degree	2
(|V|, |E|)	(3, 3)
Degree distribution	{2, 2, 2}
Isomorphism	

Anonymity measures



|V|: ~100 - 3M |E|: ~100 - 18M

Imprecise information but looking further
is more effective!



Q1: How to **measure** anonymity?

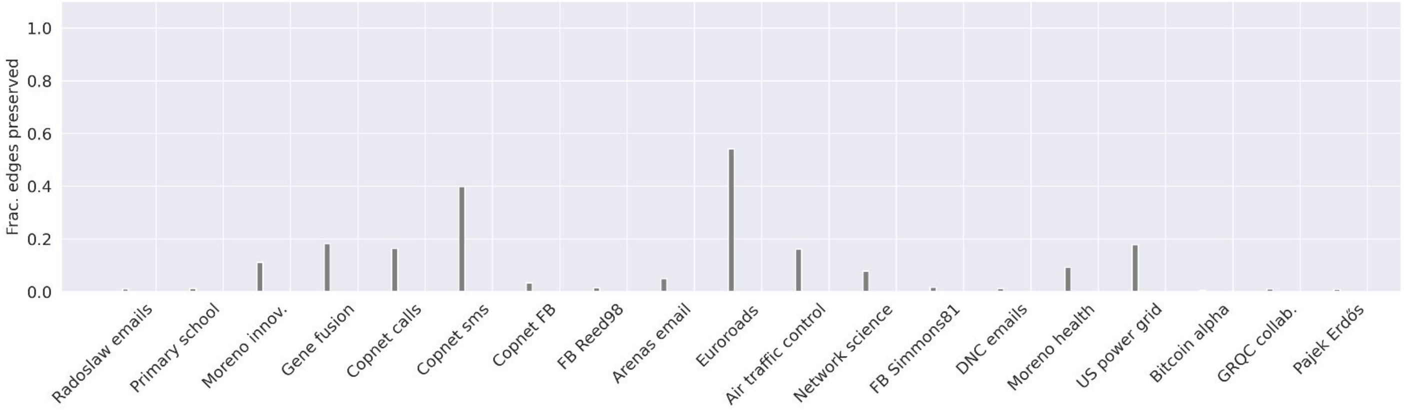
Q2: How to anonymize a network?

Q3: What is their effect on **utility**?

Deleting few edges → Higher utility(?)

Anonymization

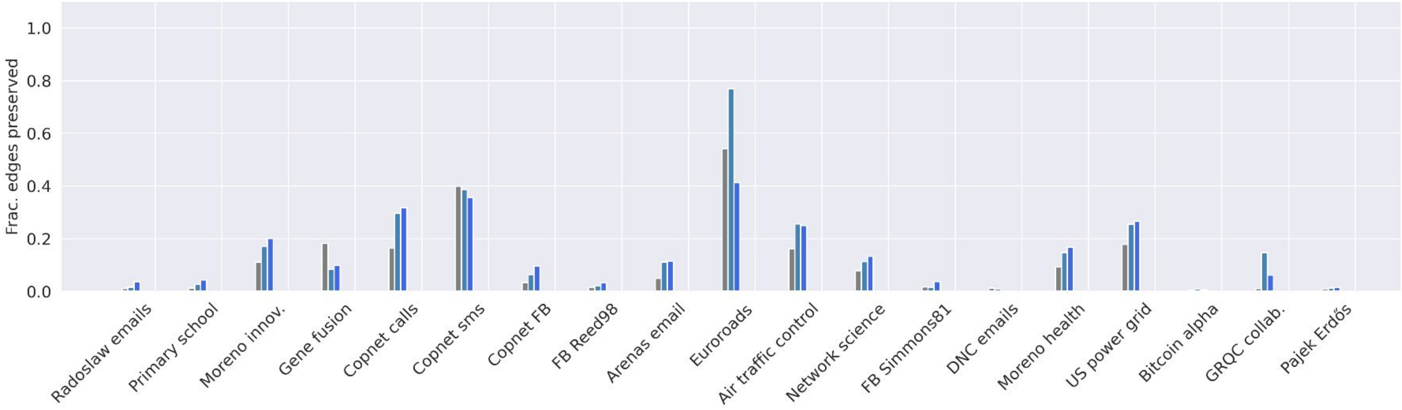
Random

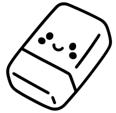


|V|: ~100 - 6,927 |E|: ~300 - 11,850

For more details see our preprint!

Anonymization





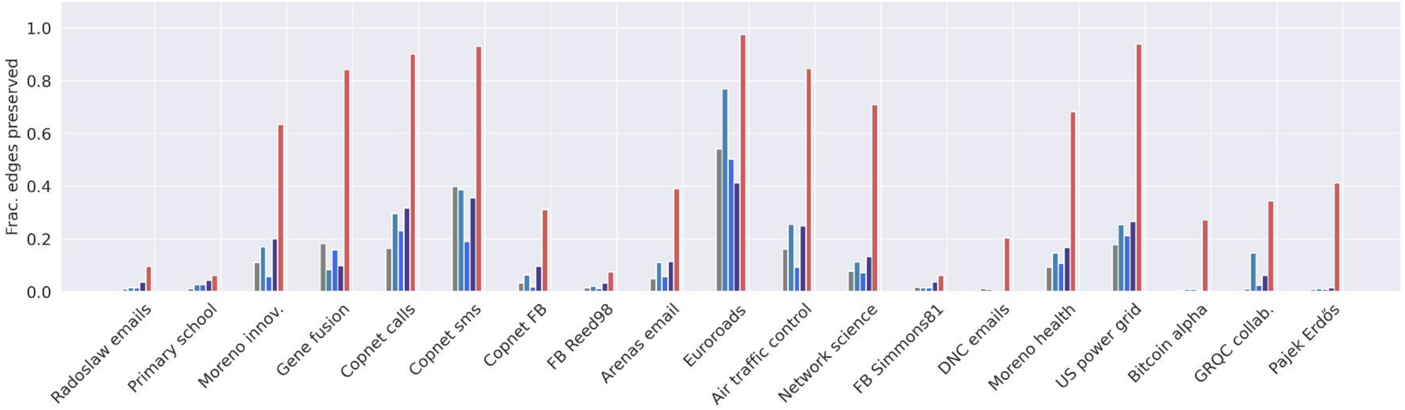
Random

Structure based

|V|: ~100 - 6,927 |E|: ~300 - 11,850

For more details see our preprint!

Anonymization



Random

Structure based

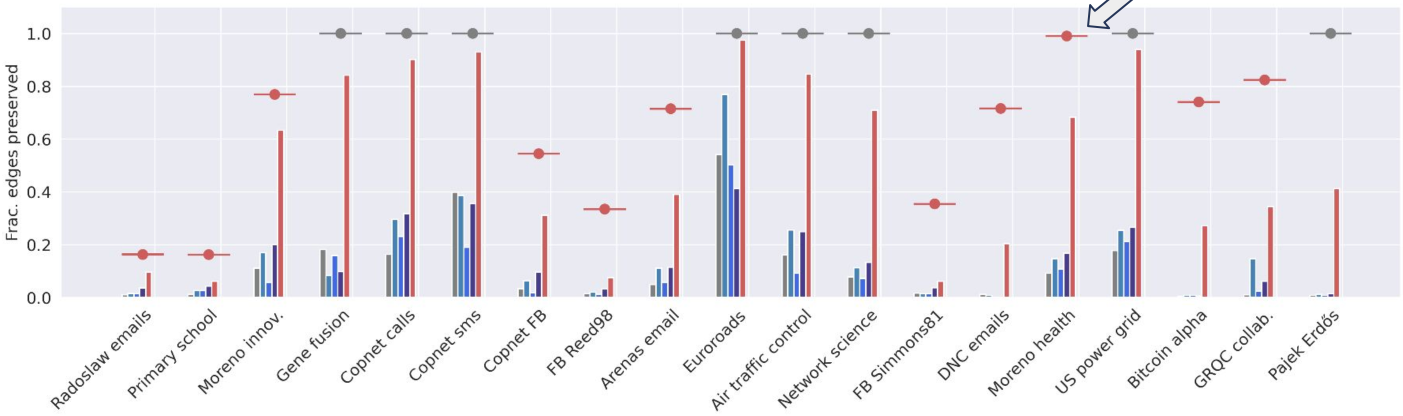
Uniqueness based

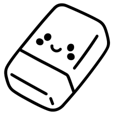
|V|: ~100 - 6,927 |E|: ~300 - 11,850

For more details see our preprint!

Anonymization

95% anonymous





Random

Structure based

Uniqueness based

|V|: ~100 - 6,927 |E|: ~300 - 11,850

For more details see our preprint!

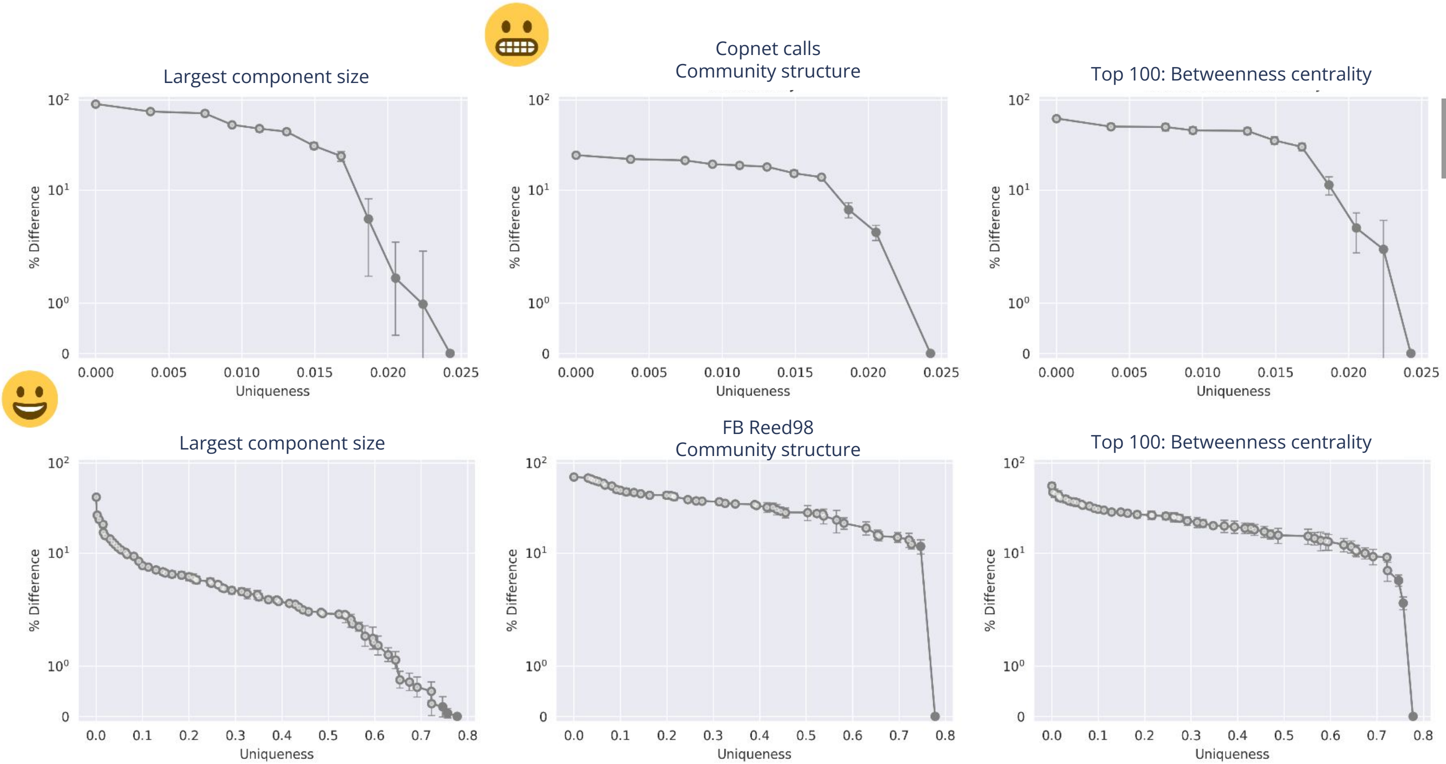
Q1: How to **measure** anonymity?

Q2: How to **anonymize** a network?

Q3: What is their effect on utility?

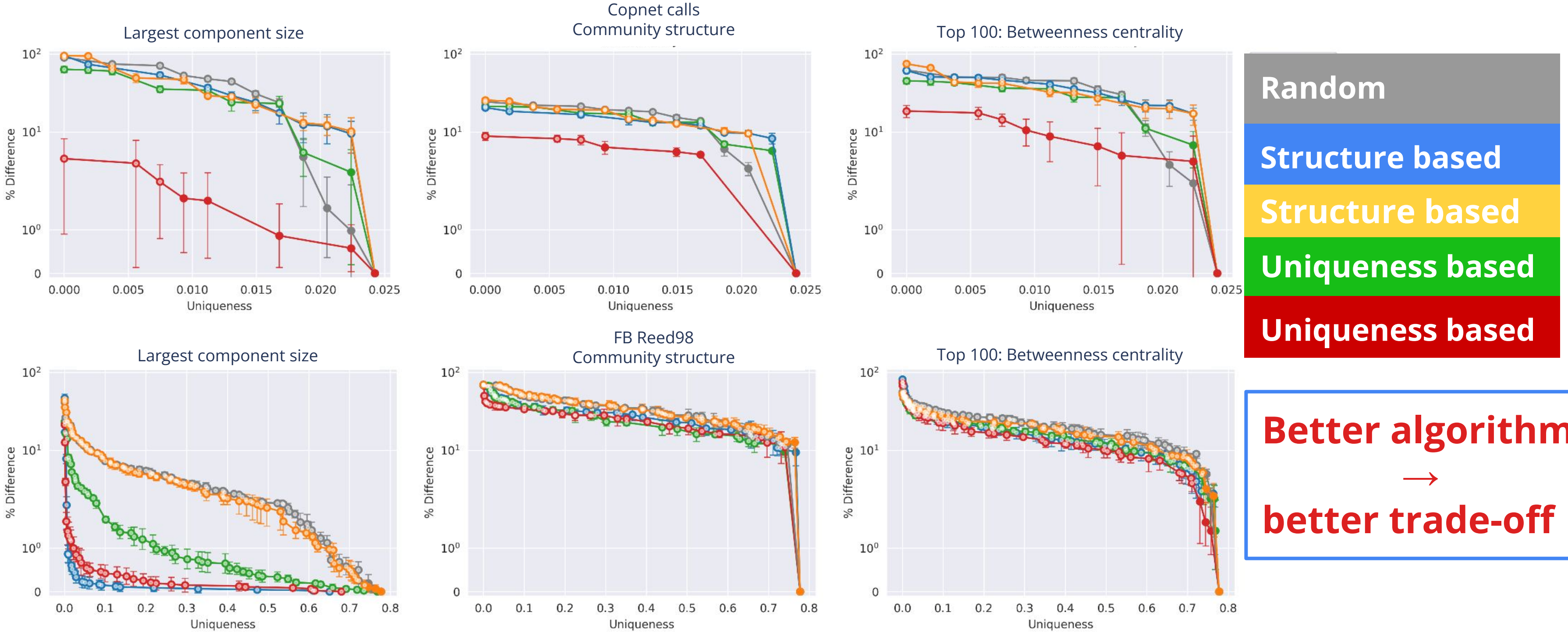
- Algorithms
- Measures

Anonymity vs. utility: **algorithms**



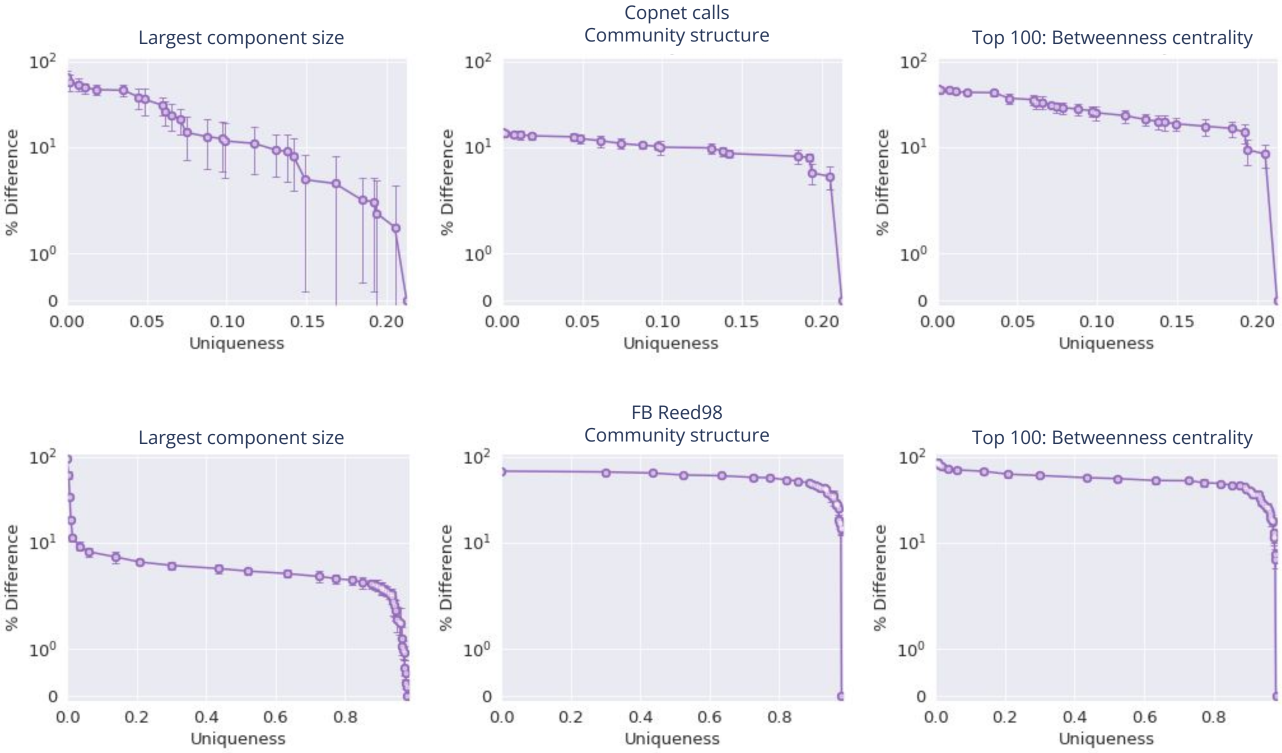
Random

Anonymity vs. utility: **algorithms**



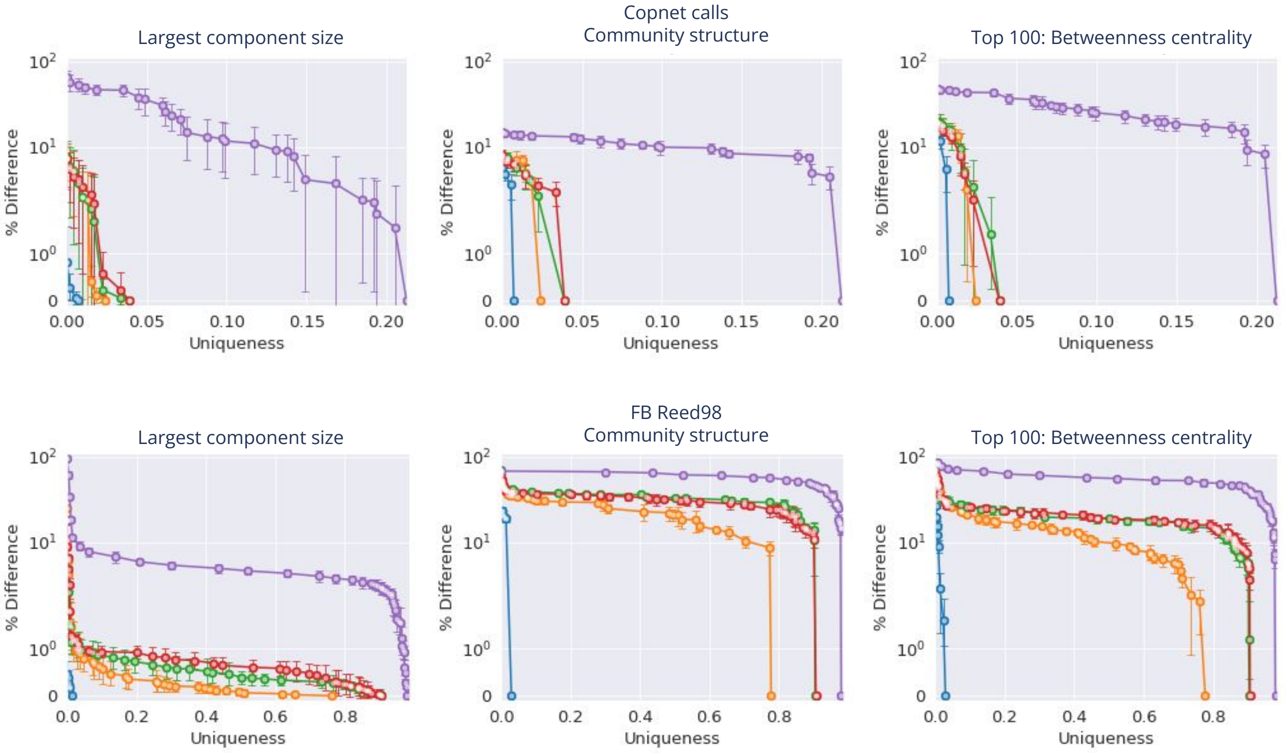
Better algorithm → better trade-off

Anonymity vs. utility: **measures**



Degrees of neighbors

Anonymity vs. utility: **measures**



Degrees of neighbors

Isomorphism

Degree distribution

(|V|, |E|)

Degree

More strict measure
→
worse trade-off

Conclusion

Q1: Measuring anonymity

- Measure = attacker scenario
- **Imprecise information and looking further** is more effective

Q2: Anonymization

- Account for **which nodes are unique!**

Q3: Utility vs. anonymity:

- Stricter measure → worse trade-offs
- Better anonymization **algorithms** → better trade-offs

Much more to do!

- Design better **anonymization algorithms**
  - Preserving certain **utility** metrics
- Design anonymity **measures** accounting for
  - Timestamps, node / edge labels

Rachel de Jong

Email: r.g.de.jong@liacs.nl

ANONET package: implement and test your own anonymization algorithms!

github.com/RacheldeJong/ANONET

[1] **The effect of distant connections on node anonymity in complex networks.** Scientific Reports 14(1), 1156 (2024)
[2] **Algorithms for efficiently computing structural anonymity in complex networks.** ACM Journal of Experimental Algorithmics 28 (2023)
[3] **A systematic comparison of measures for k-anonymity in networks** (2024) arXiv:2407.02290
[4] **The anonymization problem in social networks**, preprint arXiv:2409.16163 (2024).
Authors: R.G. de Jong, M. P. J. van der Loo, F. W. Takes

CNS group at NetSci

Monday	15:15 – 15:30		Gamal Adel	Community Structure Dynamics in a Population-Scale Multilayer Social Network	Contributed
	15:20 – 15:40		Alberto Ceria	Higher-order temporal network prediction and interpretation	Contributed
Tuesday	11:50 – 12:00		Fabrizio Corriera	Fairness in Community Detection	Lightning
Wednesday	9:45 – 10:30		Frank Takes	Connectivity and Community Structure of Online and Register-based Social Networks	Lightning
	11:00 – 13:00		Alberto Ceria	The relevance of higher-order ties	Contributed
	14:30 – 16:00		Eszter Bokányi	Mobility and cohesion of personal networks over the lifecourse of an entire population	Contributed
	—		Shane Mannion	Fast degree-preserving rewiring of complex networks	Poster
	—		Akrati Saxena	Group Fairness Metrics for Community Detection Methods in Social Networks	Poster
Thursday	14:30 – 16:00		Rachel de Jong	Navigating the privacy vs. utility trade-off in network anonymization	Contributed
	—		Yuliia Kazmina	Pathways to Upward Mobility in a Longitudinal Population-scale Social Network	Poster

Position:
PhD Candidate in AI for Network Analysis!

SCAN ME